AggieSTAAR
Python Bootcamp

Tutorial 0:
Setting up Python

TEXAS A&M UNIVERSITY

Welcome to AggieSTAAR!

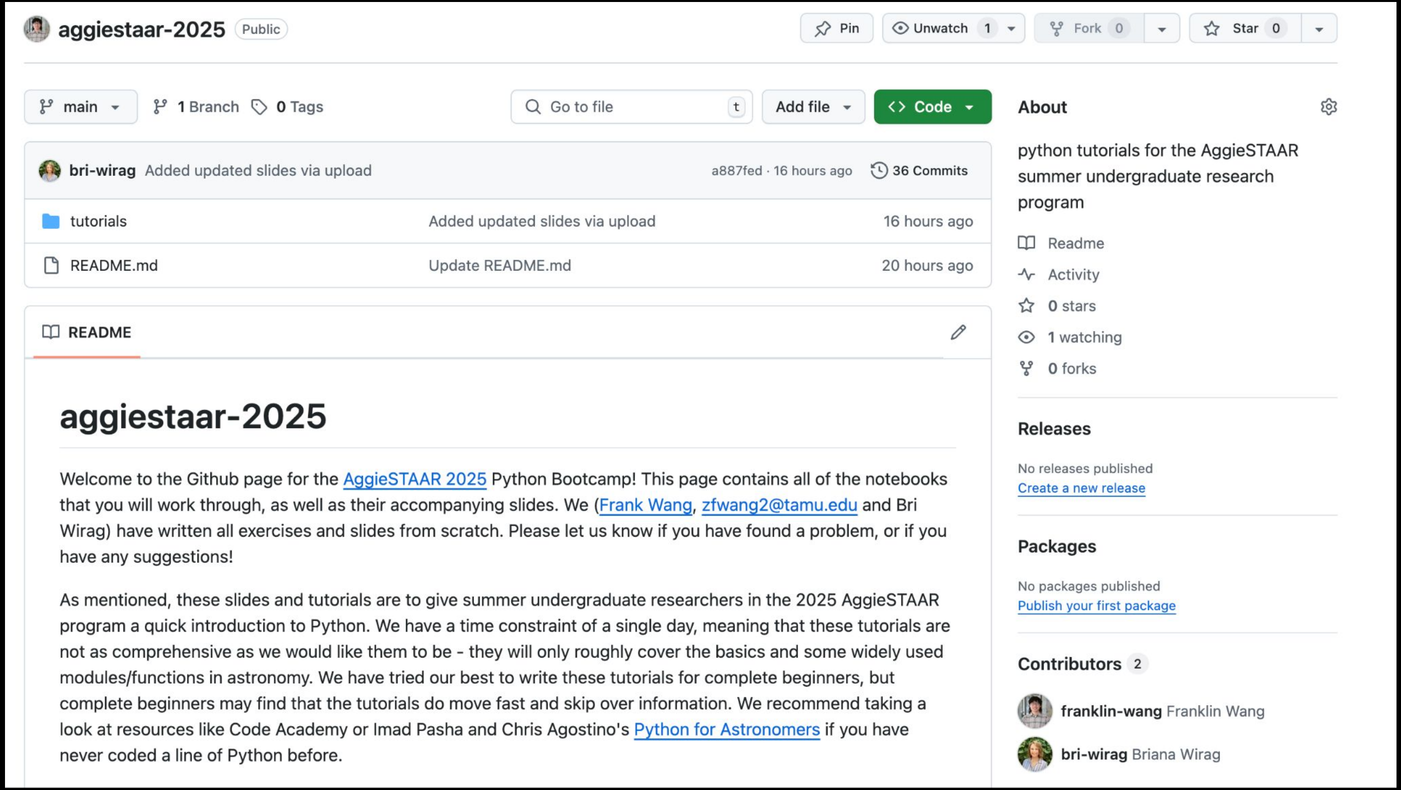
About us:

- Frank Wang, 4th year PhD, working on black holes w/ Jonelle Walsh. zfwang2@tamu.edu, office M318.
- Bri Wirag, 3rd year PhD, now working on black holes w/ Jonelle Walsh, wirag@tamu.edu, office M315.

Tell us about you!

- Name, major, year in school, and who will you working with / what you'll be working on this summer?

All of today's materials can be found at:
github.com/franklin-wang/aggiestaar-2025



Rough outline for today:

10:00AM – 10:45AM: Intro & installing Python

10:45AM – 11:30AM: Python basics & troubleshooting

11:30AM – 12:00PM: Conditions & loops

12:00PM – 01:00PM: Lunch break

01:00PM – 01:45PM: matplotlib & plotting

01:45PM – 02:30PM: astropy tables

02:30PM – 03:00PM: Resources and access

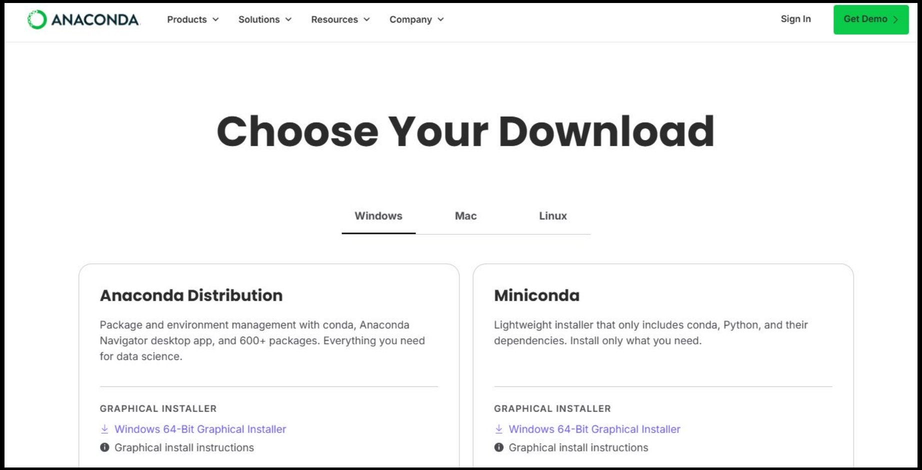
Python is your best friend!

ANACONDA colab

It is the most frequently used coding language in astronomy today. Fortunately, it's (relatively) straightforward to learn.

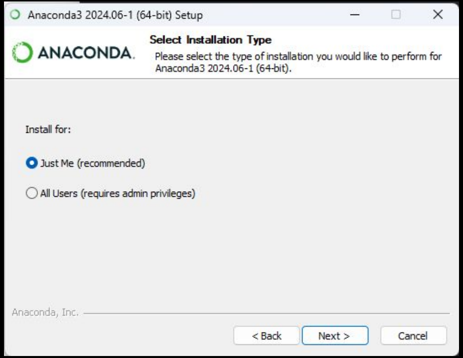
We will learn how to use Python through **notebooks**, which are user-friendly platforms that allow you to interact with code.

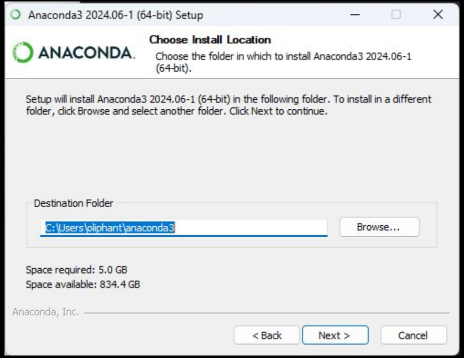
(1): Installing **Anaconda**

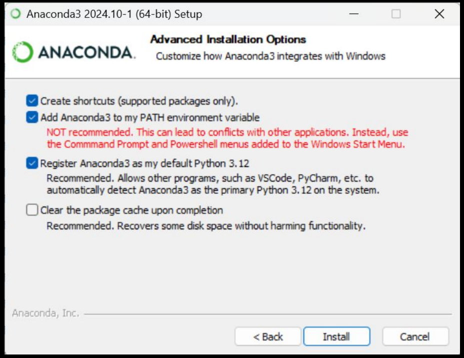


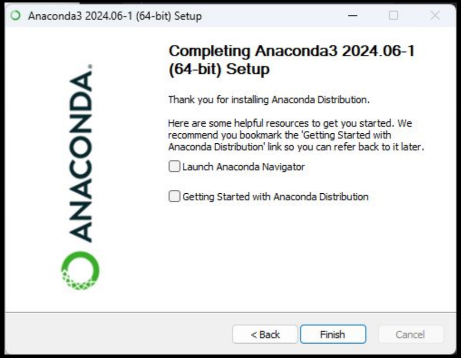
1. Download Anaconda from anaconda.com/downloads
   a. Make sure you get the correct version for your operating system (MacOS, Windows, Linux).
2. If prompted, click **Skip registration**
3. Download and run the installer

(1): Installing **Anaconda**









1. On **Select Installation Type**, choose **Just me**
2. Keep the default folder on **Choose Install Location**
3. Check the top 3 options in **Advanced Installation Options**
   a. The second option ensures that "conda" is a recognized command in terminal
4. Finish running the installer

(1): Installing **Anaconda**

Let's make sure Anaconda was installed properly, type the following into a terminal/PowerShell

```
conda env list # one item should show up
```

If an error saying "conda command not found" popped up, Anaconda was not added to your "Path" variable.

(1): Installing **Anaconda**

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Fix this by going to geeksforgeeks and follow the instructions.

- Go to “Path” under “User variables” rather than “System variables”

(2): Setting up a virtual environment

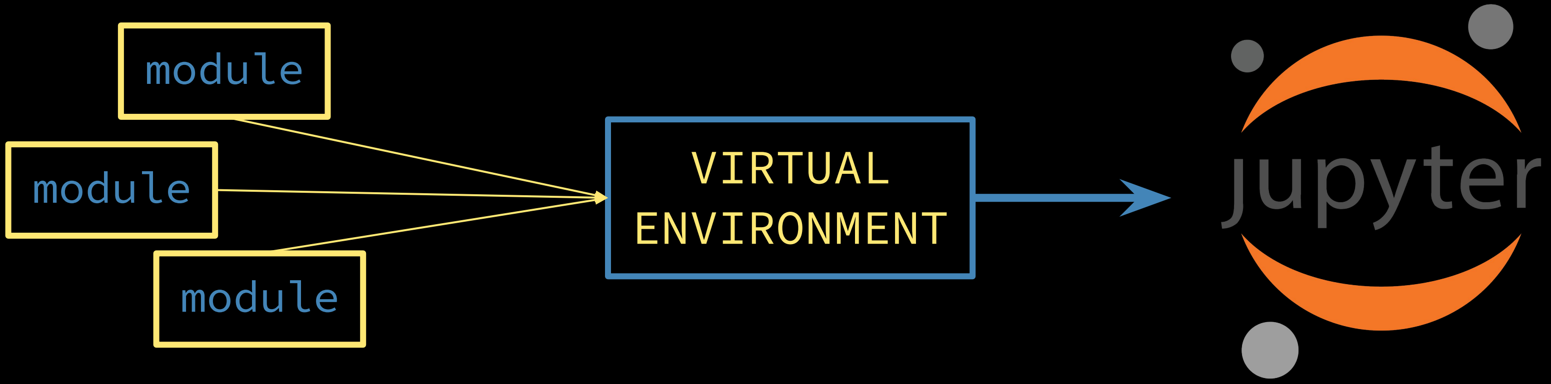
Python is a very powerful tool, but the version you install from online is pretty basic.

- Basic commands include `help()`, `print()`, `type()`, and more

Modules are what make Python useful in astronomy. If I wanted to make a plot, I would install matplotlib, since Python itself doesn't have the tools I need.

- For Jupyter to recognize and use the modules you install, one should set up a virtual environment.

(2): Setting up a virtual environment



Think of virtual environments as a toolbox, and the modules as your tools.

- You can load Jupyter without a virtual environment, but you might not have all of the tools necessary for what you want to accomplish.

If you bring your toolbox with all your custom modules, Jupyter will have a lot more functionality.

- Sometimes you need different tools, and some tools need their own toolbox, so you can have multiple virtual environments to change between and load different modules.

(2): Setting up a virtual environment

```
(base) PS C:\Users\briwi> conda env list
# conda environments:
#
base                  *  C:\Users\briwi\anaconda3

(base) PS C:\Users\briwi> conda create --name aggiestaar_env
Proceed ([y]/n)? y
(base) PS C:\Users\briwi> conda activate aggiestaar_env
(aggiestaar_env) PS C:\Users\briwi>
```

Your environment may be listed on the side here

- Open up a terminal or Powershell
- Type conda env list
  - one should show up called “base”, this is your default environment
- Type conda create --name environment_name, type “y” when prompted
  - environment_name can be (almost) anything, just make it useful
  - This creates a folder in Anaconda called “environment_name”, this will house your modules
- Type conda activate environment_name to activate your new environment!

(2): Setting up a virtual environment

Now that we've made our virtual environment , let's add some important astronomy Python modules. You can install modules two ways:

- conda install [module name] <- use this by default
- pip install [module name]   <- use if conda doesn't work

With your virtual environment activated, install the following modules using the conda command:

- numpy   (for math)
- matplotlib (for plotting)
- astropy   (a comprehensive collection of astronomy Python packages)

(2): Setting up a virtual environment

Some important notes:

- If you try to use a module and get a "module not found" error, ensure that you've:
  a) Loaded the correct virtual environment (AKA "kernel") in Jupyter
  b) Installed the module in your virtual environment
- If you are in the correct virtual environment and you're still getting the error, try reloading Jupyter.
- If the issue persists, check the module documentation, and see if there are extra installation steps you are missing.

(2): Setting up a virtual environment

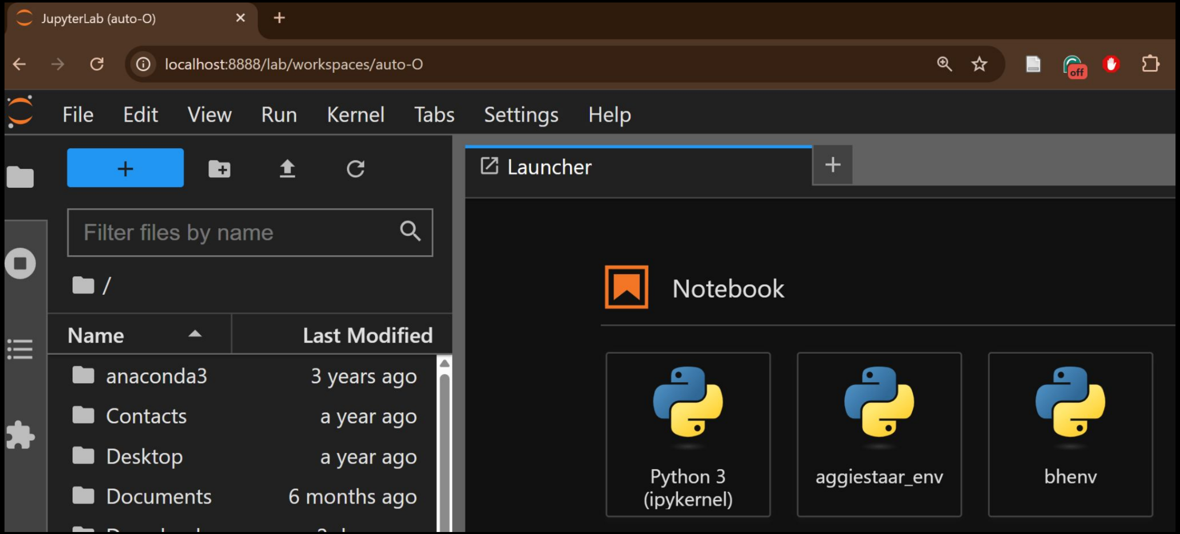
```
(aggiestaar_env) PS C:\Users\briwi> conda env list
# conda environments:
#
base                     C:\Users\briwi\anaconda3
aggiestaar_env        *  C:\Users\briwi\anaconda3\envs\aggiestaar_env

(aggiestaar_env) PS C:\Users\briwi> conda install ipykernel

(aggiestaar_env) PS C:\Users\briwi> python -m ipykernel install --user --name=aggiestaar_env
Installed kernelspec aggiestaar_env in C:\Users\briwi\AppData\Roaming\jupyter\kernels\aggiestaar_env
(aggiestaar_env) PS C:\Users\briwi> _
```

Now we need to make sure Jupyter recognizes your new virtual environment. From your terminal:

- Check that you are still in your desired environment by typing `conda env list`
  - The active environment will be marked with a *
- Type `conda install ipykernel`, type "y" when prompted, and wait
- Lastly, type:

  `python -m ipykernel install --user --name=environment_name`

(3): Launching Jupyter Lab

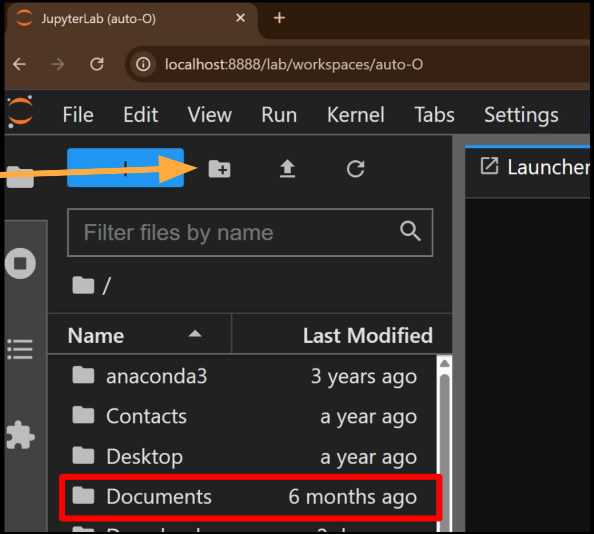


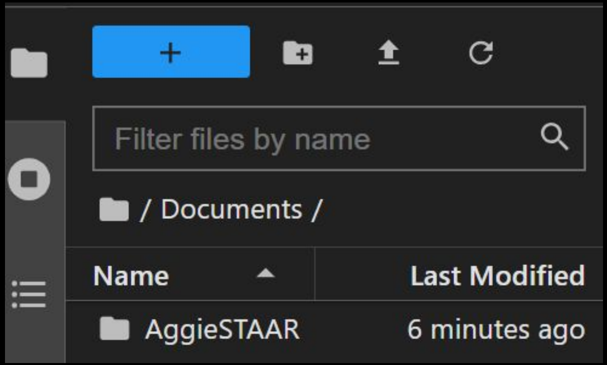
Now start up Jupyter:

- From your terminal, type jupyter lab
- Jupyter should open in a browser and look something like the picture

(3): Launching Jupyter Lab

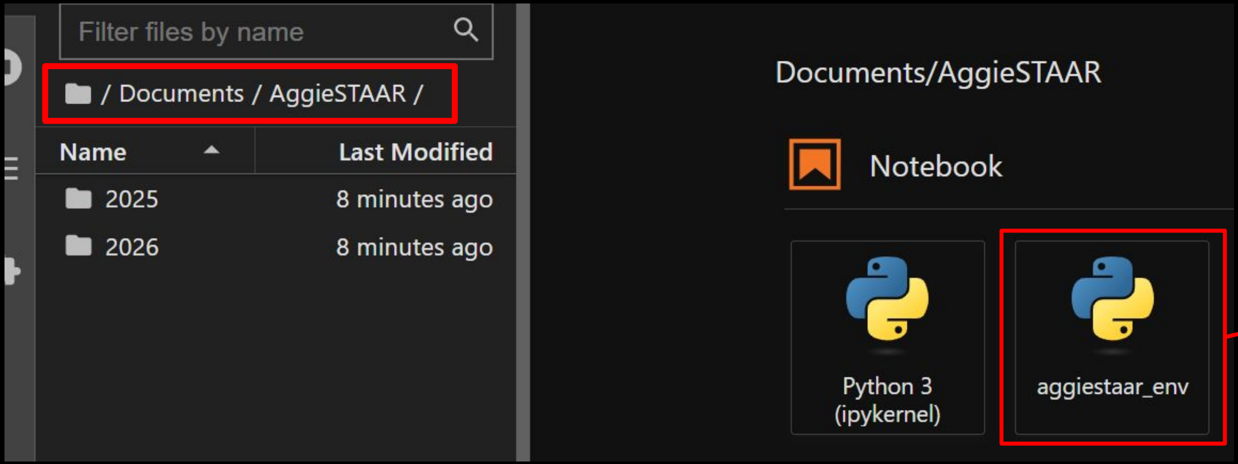
Click here for a new folder





- On the left side, you should see a file explorer
  - This is linked to your computer's file explorer, whatever you do here happens on your computer too
- Let's make a folder to store today's activities
  - Click on Documents, add a new folder, name it "AggieSTAAR"
  - Should look like the right picture

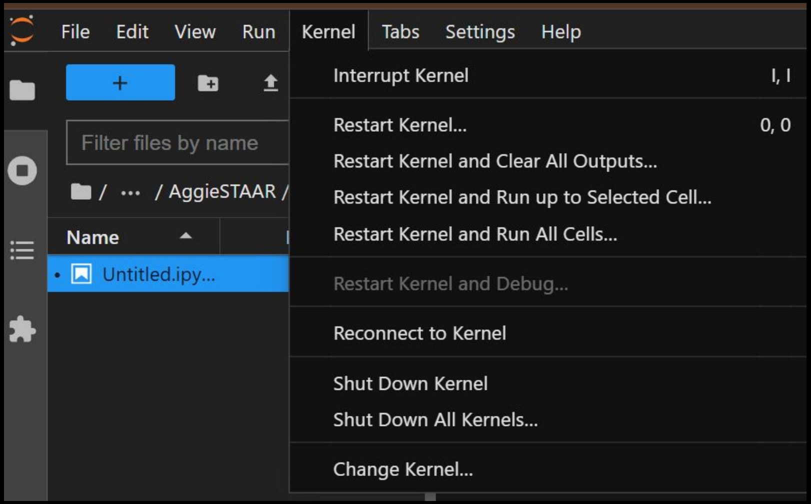
(4): Creating a Notebook

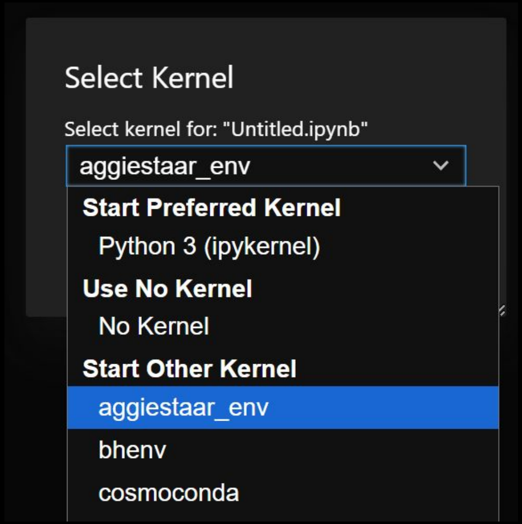


Start a new notebook by clicking here.

- Navigate to your AggieSTAAR folder (notice at the top Jupyter tells you where you are at)
- Add a new Notebook by clicking the icon with your environment name

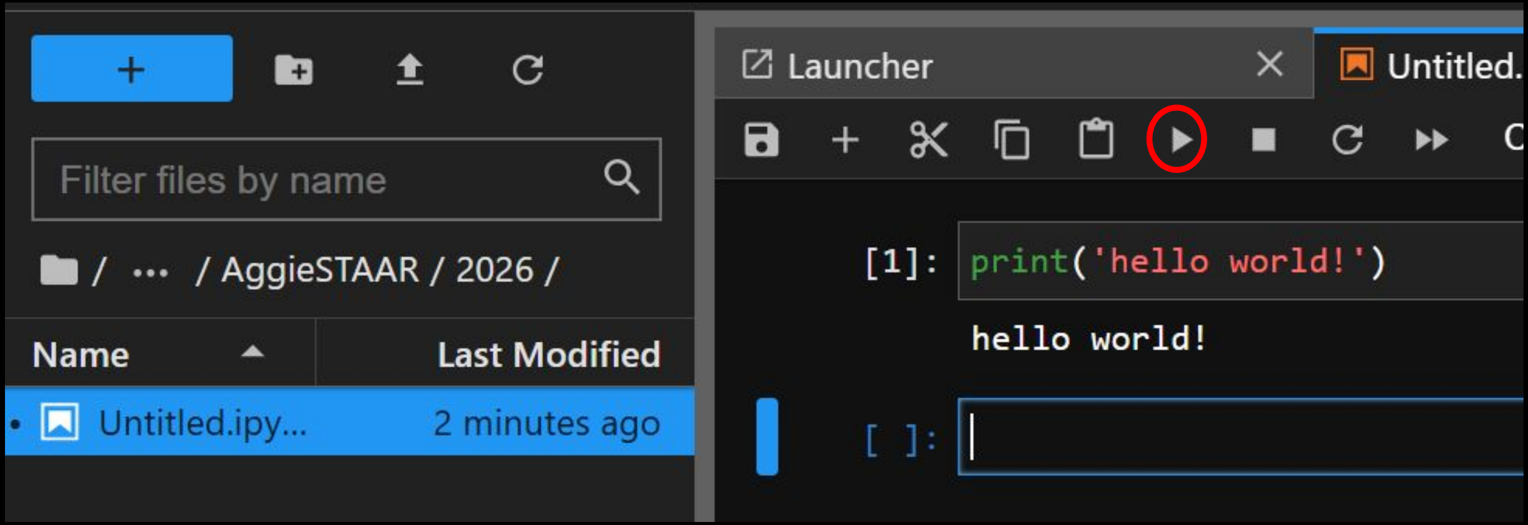
(4): Creating a Notebook





If you'd like to change the virtual environment, navigate to "Change Kernel…" under the "Kernel" tab, and select the virtual environment that you wish to use

(4): Creating a Notebook

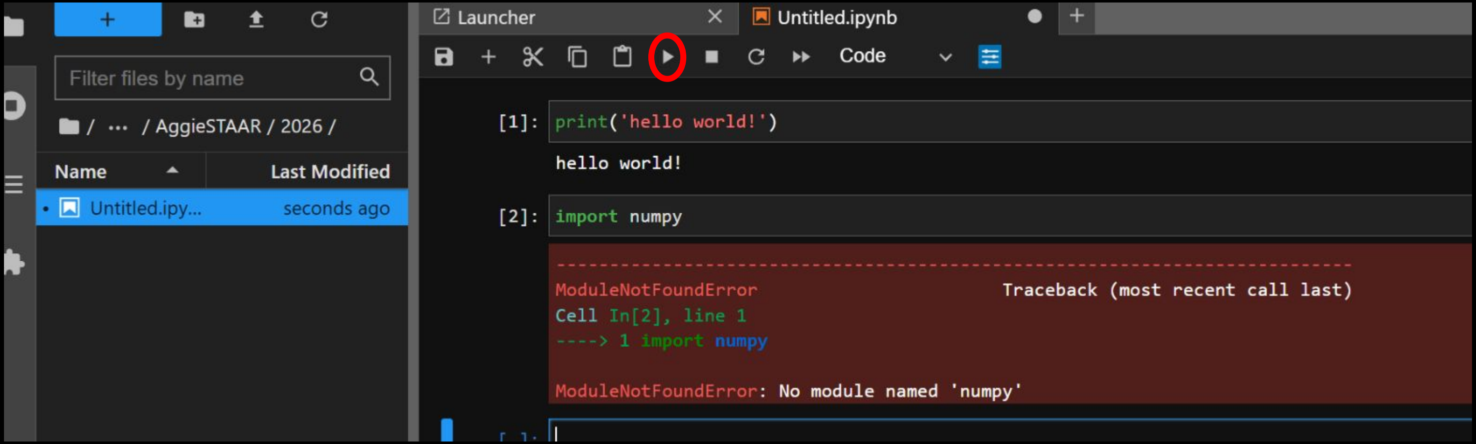


For fun, I'd like you all to type the following EXACTLY (we'll get into why later) into the first code block

- print('hello world!')
- then press SHIFT + ENTER to run the code block.

You've executed a line of python code!

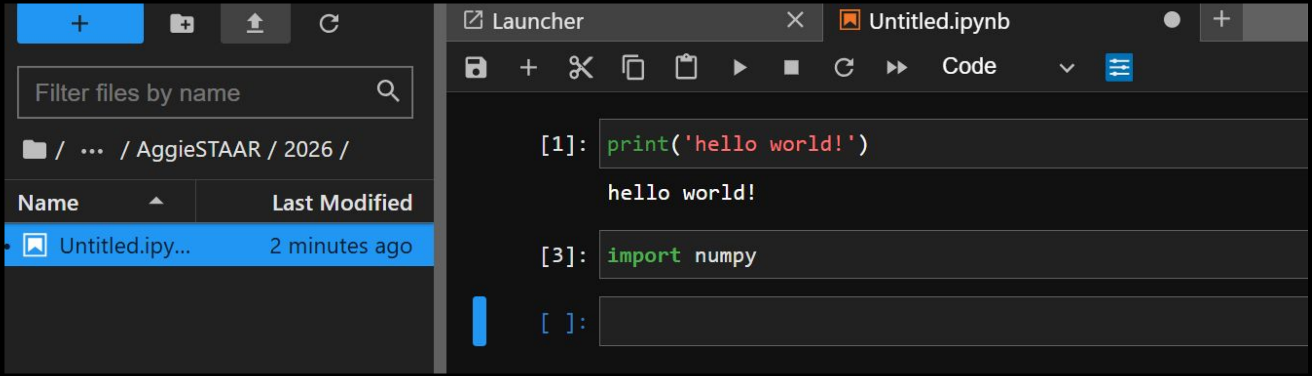
(4): Creating a Notebook



Test that everything was installed OK:

- Type import numpy into the next code block, then press SHIFT + ENTER again to execute the code block.
- If you haven't installed numpy, you'd see the above error

(4): Creating a Notebook



Since we've already installed numpy, this should run without issue!

If you run the code block and nothing happens, congratulations! You've created a Notebook and imported a module from your virtual environment.